The Project

The Minimal Viable Product:

The team's minimum viable product is a fully functional robotic arm integrated with the existing RE-RASSOR rover platform. The arm will primarily consist of 3D printed components and will utilize brushless DC motors for motion as requested by the sponsor. The arm must be capable of lifting, moving, and placing the project's 3D printed pavers while maintaining rover stability and limiting interference and changes to the existing drivetrain or chassis.

The arm will be controlled through the existing rover software architecture. Commands will be transmitted through the rover's current network-based control system and integrated into the existing web application. Software modifications will include the addition of new HTTP requests, changes to the user interface controls to make a seem less design between rover and arm, and motor control logic necessary to operate the arm.

The project will be considered complete when the arm can successfully manipulate pavers, operates through the existing rover control interface, and is able to be demonstrated on a physical rover platform. The final deliverables will include CAD files, assembly instructions, software source code, testing results, and documentation sufficient for future student teams to continue development.

The Stretch Goals:

While the primary objective of this project is to design a functional robotic arm that can lift and manipulate 3D printed pavers, our team with the guidance of the Florida Space Institute has identified several areas that would further expand the RE-RASSOR rovers' capabilities if time is sufficient and our primary objective has been completed and approved by our sponsor.

Autonomous Object Pickup

One stretch goal is the implementation of the arm to act autonomously and pick up pavers without human control. Instead of relying on manual controls, the rover would identify target objects (i.e. our 3D printed Pavers) and position the arm and the drivetrain to pick up the object. This feature would require adding more cameras to the rover specifically used for object detection, computer vision software, and routines for object detection. The system would also need to be able to convert detections into valid coordinates that it can then decide to move the rover to. Overall, this would add significant complexity to the project but would increase the education value and interest in the project for our target audience.

Additional Wrist Joint for Rotation

In our brainstorming sessions in the arm, we are looking towards having 3 degrees of freedom in our arm, with one of them going on the wrist to allow rotation in the pitch axis along the arm's body. This design neglects the chance to allow our wrist to rotate along the roll axis, decreasing its effectiveness for certain objects, such as a handle with an opening on the side. This is a significant change and complexity add to our current scope, as we would have to modify our final arm design to have a satisfactory weight, and handle the wiring to hook up a 2^{nd} Brushless DC motor for the wrist. This would also require us to spend time CADing new designs and testing them before finalizing an additional joint.

Interchangeable End Effector Attachments

Another potential stretch goal is the development of interchangeable end-effector attachments for the robotic arm. Our base deliverable will focus on one end effector design optimized for picking up 3D printed pavers, but a modular mount system would allow the rover to perform a wider variety of tasks and allows users to design their own end effectors to test. Implementing the ability for interchangeable end effectors would make the rover a more useful education tool as well, as it allows students to focus on one aspect, the end effector, to practice learning STEM concepts. Additionally, this stretch goal is much more obtainable compared to autonomous pickup or driving, since it requires us to modify already planned work in the arm design.

Confirmation of Sponsor's Approval:

Hi Christian,

I hope you're doing well. For one of our Senior Design deliverables, Professor Gerber requires written confirmation from our sponsor that our proposed project scope and Minimum Viable Product (MVP) align with the sponsor's expectations.

I've attached a summary of our proposed MVP and identified stretch goals based on our discussions and our current understanding of the project. Could you please review the attached document and let us know if this accurately reflects the scope and objectives you expect our team to complete?

If it does, a simple reply confirming that the attached plan matches your expectations would satisfy our assignment requirements. If there are any changes or corrections you would like us to make, please let us know and we will update the document accordingly.

Thank you sincerely,

Kaitlyn Armstrong

CV Christian Vincent

To: Kaitlyn Armstrong

Reply Reply all Forward

Thu 2026-06-25 3:37 PM

Retention: UCF Delete after 10 Years (10 years) Expires: Sun 2036-06-22 3:37 PM

Looks good.

-Christian Vincent

Our Current Accomplishments:

Since the project's assignment, the team has completed several foundational tasks necessary for future development. The team participated in sponsoring meetings with representatives from the Florida Space Institute to gain a better understanding of project requirements, expectations, and available resources. These meetings provided valuable insight into the project's goals, previous development efforts, and technical constraints.

A significant portion of the team's initial efforts focused on reviewing previous RE-RASSOR repositories and documentation. Through this investigation, the team identified the previous brushless DC motor (BLDC) team's work as an important technical reference for future development. Reviewing this material has provided valuable information regarding previous design decisions, motor integration challenges, and potential approaches that may be applicable to the arm development project.

The team has also completed several organizational and planning tasks. These accomplishments include the creation of the Team Contract, development of the Design Proposal, and assignment of team roles and responsibilities. Completing these deliverables established expectations for communication, workload distribution, and project management while also providing a foundation for future design decisions.

In addition to project planning activities, team members have begun technical preparation for future development. Initial research has been conducted on CAD software, 3D printing workflows, available manufacturing resources, and the capabilities of the Bambu Lab X1 Carbon printers available through the Florida Space Institute. This research is intended to support the design and manufacturing of prototype arm components later in the semester.

The team has also begun investigating the rover's existing software architecture and control systems. Initial efforts have focused on understanding how commands are transmitted through the rover web application, the role of the Raspberry Pi in processing those requests, and how future arm controls may integrate into the existing software framework. This research will help ensure that future software development remains compatible with the current RE-RASSOR platform.

Lastly, the GitHub repository for our project has been fully set up with proper credits to the previous team's work. There are 3 file branches: the first specifically for our work on the rover, the second for previous team's work that we will directly be incorporating into our work, and the third is for all documentation we generate over the course of this class. This file structure, plus the through ReadMe has been constructed to afford proper credits and set the basis for documentation moving forward. In order to ensure all previous directly implicated artifacts were included in the repository, git Large File Storage (LFS) was added to ensure the project could be properly built upon.

<u>Our Current Research & Work:</u>

A significant portion of the team's research efforts has focused on understanding the existing RE-RASSOR platform and identifying previous work that can be leveraged throughout development. Rather than developing an entirely new rover, our team is inheriting several years of previous student work. Because of this, understanding the decisions, successes, and failures of previous teams is critical to ensuring that development time is spent improving the platform rather than recreating functionality that already exists.

One major area of investigation has been the previous Brushless DC Motor (BLDC) project completed by an earlier Senior Design team. The sponsor has requested that the robotic arm also utilize BLDC motors, making the previous team's work directly relevant to our project. Through reviewing their documentation and repositories, we have researched motor selection, motor control methods, gearbox design, and software libraries used for motor control. This work has provided valuable information regarding both successful approaches and challenges that may influence future arm development.

The team has also researched the rover's existing software architecture. Initial investigation has focused on understanding how commands are transmitted through the rover's web application, how those commands are processed by the Raspberry Pi, and how future arm controls can be integrated into the current system. Since one of the project requirements is maintaining compatibility with the existing rover interface, understanding this architecture is an important prerequisite for future software development.

Additional research has focused on CAD software, additive manufacturing workflows, and available fabrication resources. Team members have begun learning the CAD tools and 3D printing workflows necessary to produce prototype components. Research has also been conducted regarding the capabilities of the Bambu Lab X1 Carbon printers available through the Florida Space Institute, including available materials, manufacturing constraints, and best practices for producing functional mechanical components.

The current workload of the back-end team consists of working on testing existing http requests to ensure that they understand how the functions work within the existing code in order to replicate those processes in the software of the arm. Preliminary tests are being conducted, and progress is being logged in our team’s folder of work within the GitHub repository. Meanwhile, the front-end team is working on downloading and editing the existing rover to add the base of the arm prototype. We are formulating plans for test prints test prints to ensure that when printing the larger more materially expensive components of the robot, we reduce the odds of running into common print errors that could lead to misprints that can occur under improper settings that would result in an unstable product.

<u>What We Need to Research:</u>

Although significant progress has been made in understanding the existing platform, several areas require additional investigation before prototype manufacturing can begin. One of the team's primary research goals is determining the final mechanical requirements of the arm. This includes identifying required payload capacity, expected range of motion, stability constraints, and acceptable operating limits when mounted to the rover.

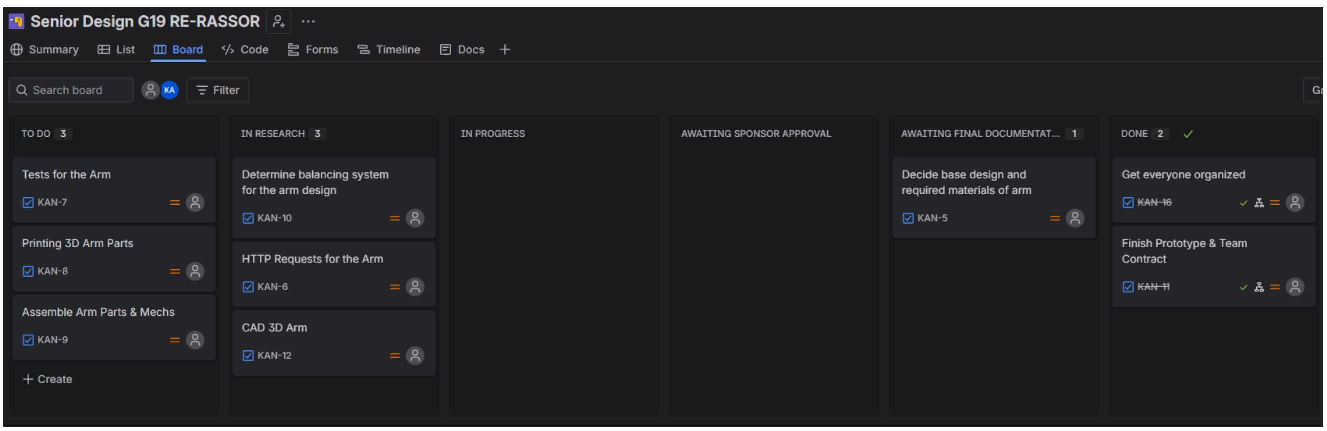
The team must also continue researching motor requirements and drivetrain solutions. While previous teams have successfully implemented BLDC motors within the rover platform, the loading conditions and motion requirements of a robotic arm differ significantly from those of a drivetrain. Additional research will therefore be required to determine whether existing gearbox designs can be reused directly or if modifications are necessary.

Software integration remains another important area of future research. While the team has begun investigating the rover's command structure and HTTP-based control system, additional work is needed to determine how arm controls should be incorporated into the existing web interface. The team will also need to investigate communication reliability, command structures, safety considerations, and future maintainability.

Finally, the team will continue researching arm design methodologies, manufacturing approaches, and testing procedures. Since the project will likely require multiple design iterations before a final solution is reached, establishing repeatable testing methods and evaluation criteria will be important for guiding future design decisions.

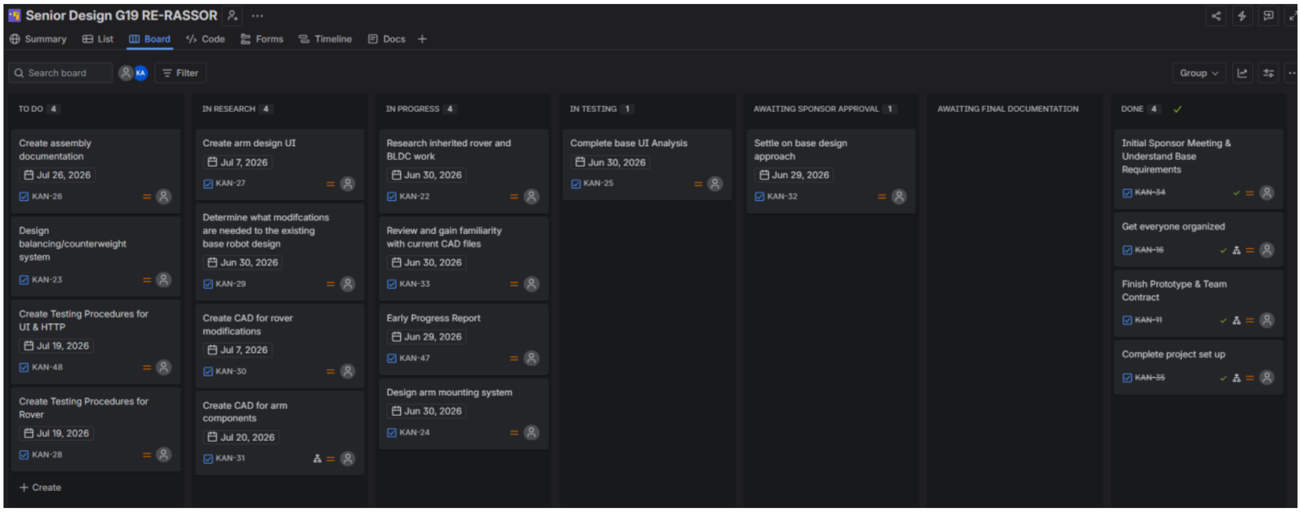
The Timeline

Initial Kanban Logs



The initial Kanban board was created to organize tasks identified during the team's research and planning phase. Because the project is based on an inherited rover platform, the initial backlog focused primarily on understanding previous work, identifying technical requirements, learning required technologies, and establishing project infrastructure. Tasks were divided into research, development, sponsor review, documentation, and completion stages to provide visibility into project progress while maintaining flexibility for future design changes. As the team gains additional information and prototype testing results become available, the board will continue evolving to reflect current project priorities.

Current KanBan Logs



Since the creation of the initial Kanban board, several tasks have progressed from project planning and research into active development. The team has completed foundational project setup activities, including sponsor meetings, team organization, project documentation, repository organization, communication infrastructure, and the selection of an initial arm design. These completed tasks have provided the framework necessary to begin prototype development and technical implementation.

The current Kanban board is organized to reflect the project's transition from research into engineering development. Active work includes reviewing inherited rover and BLDC motor resources, becoming familiar with existing CAD models, investigating the rover's software architecture and HTTP request structure, and evaluating modifications required to integrate the arm onto the existing rover platform. The team is also working to identify structural limitations of the current rover design and determine what changes may be required to support the arm safely and effectively.

Current backlog items include designing a balancing and counterweight system to address rover stability concerns, creating CAD models for arm components and rover modifications, developing the user interface required for arm operation, and establishing assembly and testing procedures. Additional work will focus on determining the optimal mounting solution, validating motor and gearbox integration, and preparing components for prototype manufacturing.

As development progresses, tasks will continue moving through research, implementation, testing, sponsor review, and final documentation stages. The Kanban board will be updated regularly to reflect project priorities, prototype results, and sponsor feedback. Because the project follows an iterative engineering process, future tasks and priorities may evolve as testing identifies new requirements, design limitations, or opportunities for improvement.

Notional KanBan Plan

Week of June 22 - June 28

- Complete Early Status Report.
- Review existing arm CAD model.
- Familiarize team members with the current design.
- Identify potential design weaknesses and areas requiring modification.
- Continue investigating rover software architecture and HTTP request structure.
- Continue CAD and 3D printing skill development.
- Begin documenting required modifications to the existing design.
- Deliverable:
    - Team understanding of inherited arm design and identified improvement areas.

Week of June 29 - July 5

- Modify existing CAD models.
- Evaluate mounting strategy on the current rover platform.
- Review motor placement and gearbox integration.
- Determine manufacturing requirements.
- Review component compatibility with available materials and printers.
- Continue software architecture investigation.
- Deliverable:
    - Revised Version 1 CAD model.

Week of July 6 - July 12

- Finalize initial design modifications.
- Complete CAD updates for prototype manufacturing.
- Generate print-ready files.
- Review the assembly process.
- Identify required purchased hardware and fasteners.
- Deliverable:
    - Prototype-ready CAD package.

Week of July 13 - July 19

- Begin manufacturing prototype components.
- Print structural arm components.

- Assemble printed components.
- Resolve assembly issues as they are discovered.
- Begin preliminary testing of arm movement.
- Deliverable:
    - Partially assembled the prototype.

Week of July 20 - July 26

- Complete assembly of prototype arm.
- Integrate motors and mechanical systems.
- Mount prototype to the rover.
- Begin testing arm functionality.
- Identify design failures and improvement opportunities.
- Deliverable:
    - Fully assembled prototype.

Week of July 27 - July 31

- Conduct prototype testing.
- Evaluate lifting capability.
- Evaluate stability and reliability.
- Document performance.
- Gather sponsor feedback.
- Create Fall redesign backlog.
- Deliverables:
    - Functional arm prototype demonstrated before the end of summer semester.
    - Documented redesign recommendations for Fall development.

The project timeline is organized around a Kanban workflow rather than fixed Scrum sprints. Because the project involves significant prototyping, testing, and redesign, work items are expected to move dynamically between research, development, testing, and documentation stages. The primary Summer 2026 objective is the creation of a functional prototype, while the Fall 2026 semester will focus on refinement, integration, testing, and final delivery.

Based on the results from our initial prototype, we will construct a complete kanban structure based off where our design needs improvement. As our project is experimental and iterative, a full kanban plan is difficult to finalize at this point. If the prototype works, it will be lightly polished and stretch goals will be analyzed. If it fails and needs significant improvements, we will analyze failure points and based off of that information reenter a shortened research and design process to produce the final model.

Currently Anticipated Risks & Challenges

While the team has established a preliminary timeline for prototype development, several technical challenges may affect the exact pace of progress throughout the project. One of the most significant risks involves rover stability and weight distribution. Previous teams encountered balance-related issues, and the addition of a robotic arm introduces further concerns regarding center of gravity and load transfer. The team will need to investigate balancing systems and potentially develop counterweight solutions to ensure the rover remains stable while operating within simulated regolith pits. Additional design and testing cycles may be required if stability issues are identified during prototype validation.

Another anticipated challenge involves integration with the existing rover platform. Preliminary investigation suggests that portions of the current rover structure may be too small or insufficiently reinforced to support the team's current arm design. As a result, modifications to the existing platform or the development of supplemental mounting structures may be required. If significant structural changes become necessary, manufacturing and testing timelines may need to be adjusted accordingly.

The team also anticipates challenges related to motor selection, gearbox integration, and arm actuation. Although previous teams have successfully implemented Brushless DC motors on the rover platform, the loading conditions associated with a robotic arm are substantially different from those of a drivetrain. Additional testing may reveal the need for redesigns or modifications to the mechanical system, which could introduce additional development time.

Because the project follows an iterative engineering process, the prototype developed during the summer semester is expected to serve as a proof of concept rather than a final design. The team has intentionally adopted a Kanban workflow to provide flexibility for redesigns resulting from testing, sponsor feedback, or newly identified technical constraints. Any issues discovered during prototype manufacturing or validation will be incorporated into the Fall development backlog and addressed through subsequent design iterations.